上海交通大学
SHANGHAI JIAO TONG UNIVERSITY

Carbon Accounting

Lecture 11

Carbon Accounting: Project/Policy

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Learning Objectives

1. **Explain** how consequential accounting estimates GHG emission changes caused by a decision.
2. **Apply** consequential accounting to a project or policy scenario by identifying the baseline and relevant emission effects.
3. **Evaluate** which accounting approach is appropriate for a given purpose, considering the accounting question, key assumptions, and limitations.

目 录

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01- Consequential Approach

02-Project/Policy Level Accounting

03-Pros and Cons of Consequential Accounting

04-Wrap-up and Next Time



Can we trust this claim?

A new offshore wind farm supplies electricity to the regional grid. The project developer claims:

"Our wind farm reduces CO_2 emissions because it generates electricity without burning fossil fuels."

What would you need to know before accepting this claim?

Think

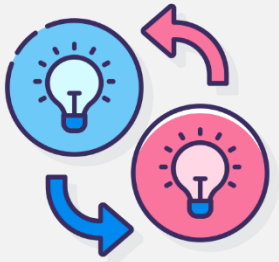
1 min



Write down **ONE** question you would ask the developer.

Pair

2 min



Exchange ideas with your neighbor and agree on **ONE** question.

Share

1 min

Yuketang Posting



Share your agreed-upon question in Yuketang by posting.

01

Consequential Approach

Background and characteristics of consequential accounting

- Term “consequential” was first used in life cycle assessment.
- Contrasted with “attributional” accounts
- Key characteristic of consequential methods:
  - Quantifies the **total change** in emissions caused by a decision or action.

system-wide consequences

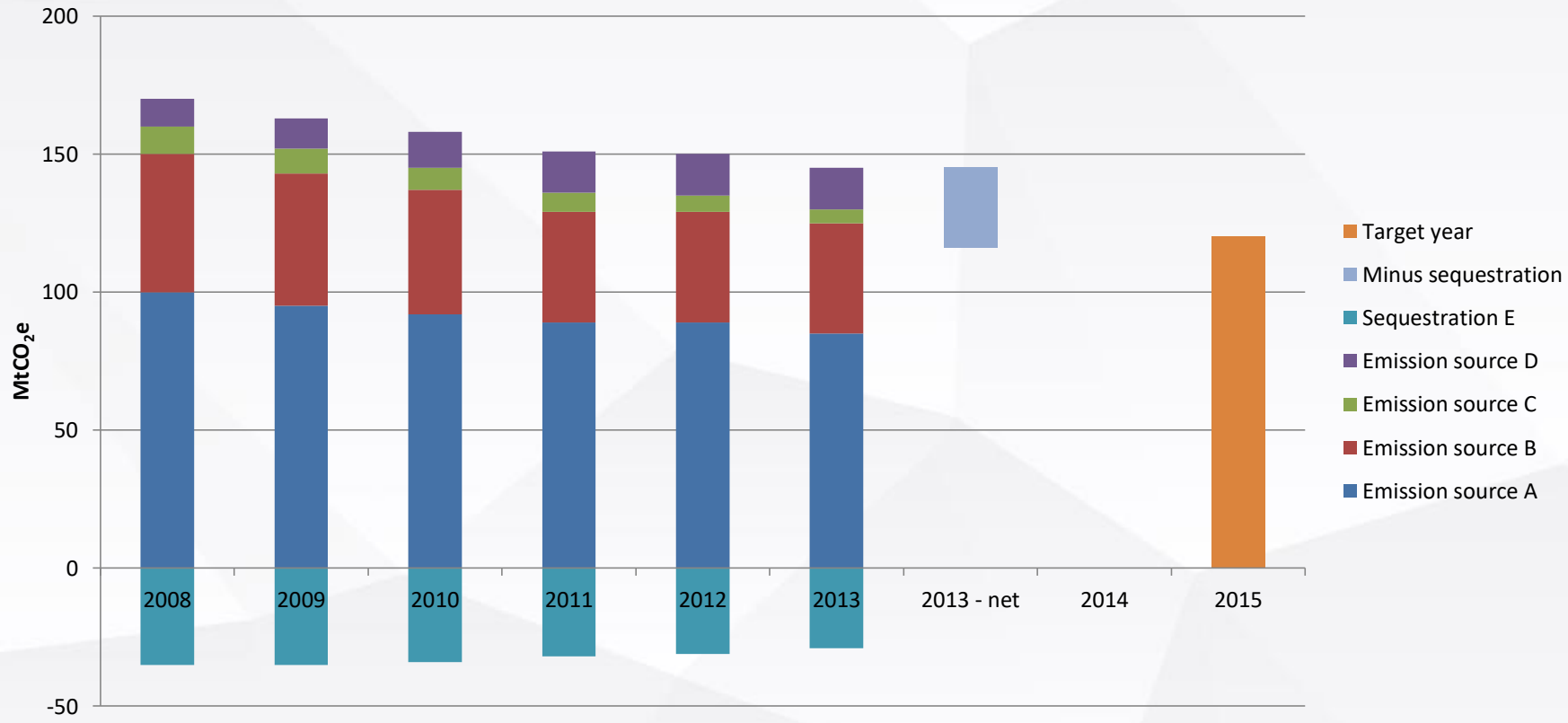


direct effect substitution effect

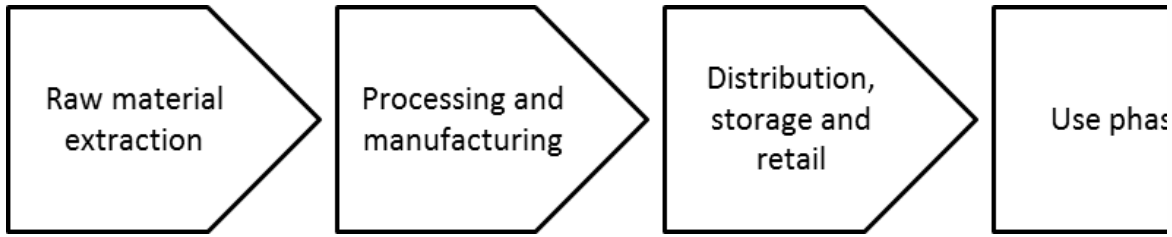
market response displacement

…

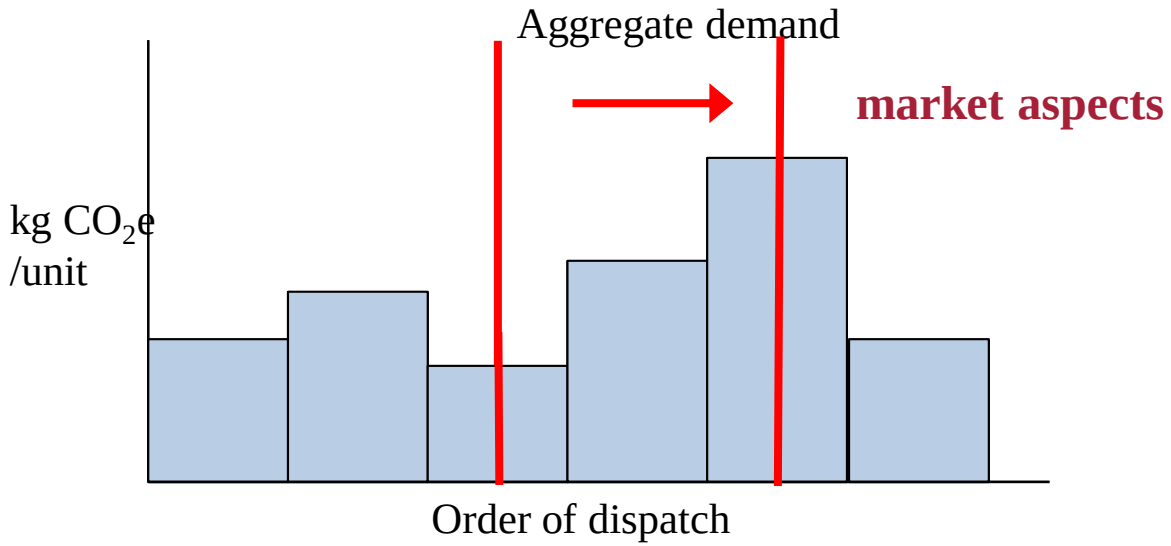
Attributional accounting



Consequential life cycle assessment (*)



- Weidema 1993 – the marginal technology the consequences for the environment.



Marginal technology is taken into use or taken out of use if the produced amounts are increased or reduced, respectively. It is not always the worst technology that is taken out of use nor modern technology that is taken into use. This is, among other reasons, due to geographical, technical, political or economic limitations in the availability of modern technology. To use marginal technology as a basis for comparison thus implies the use of either the worst, the average or modern technology, depending on which of these will actually be affected by the changes in the produced amounts. Therefore, marginal technology can give the most correct description of the actual consequences for the environment. However, in a given situation it may be difficult to establish what is actually the marginal technology, since this will depend on the above-mentioned particular geographical, technical, political and economic conditions.

(Source: Weidema, B., 1993. Market aspects in product life cycle inventory methodology. *Journal of Cleaner Production*, 1(3), pp.161–166.)

Consequential life cycle assessment - example

Table 3 Characterised results of the average conventional milk production system using attributional LCA and mass, economic allocation, besides consequential LCA and system expansion

Methodology Handling co-products impact category	Unit/kg FPCM	Attributional mass allocation	Attributional economic allocation	Consequential system expansion
Land use[a]	m^2	1.18	1.16	0.90
Fossil energy use[b]	MJ	5.77	6.91	2.55
Eutrophication[a]	g NO_3-eq	163	170	113
Acidification[a]	g SO_2-eq	10.9	11.2	4.78
Climate change[a]	g CO_2-eq	1,560	1,610	901

(Source: Thomassen, M.A. et al., 2008. Attributional and consequential LCA of milk production. *The International Journal of Life Cycle Assessment*, 13(4), pp.339–349.)

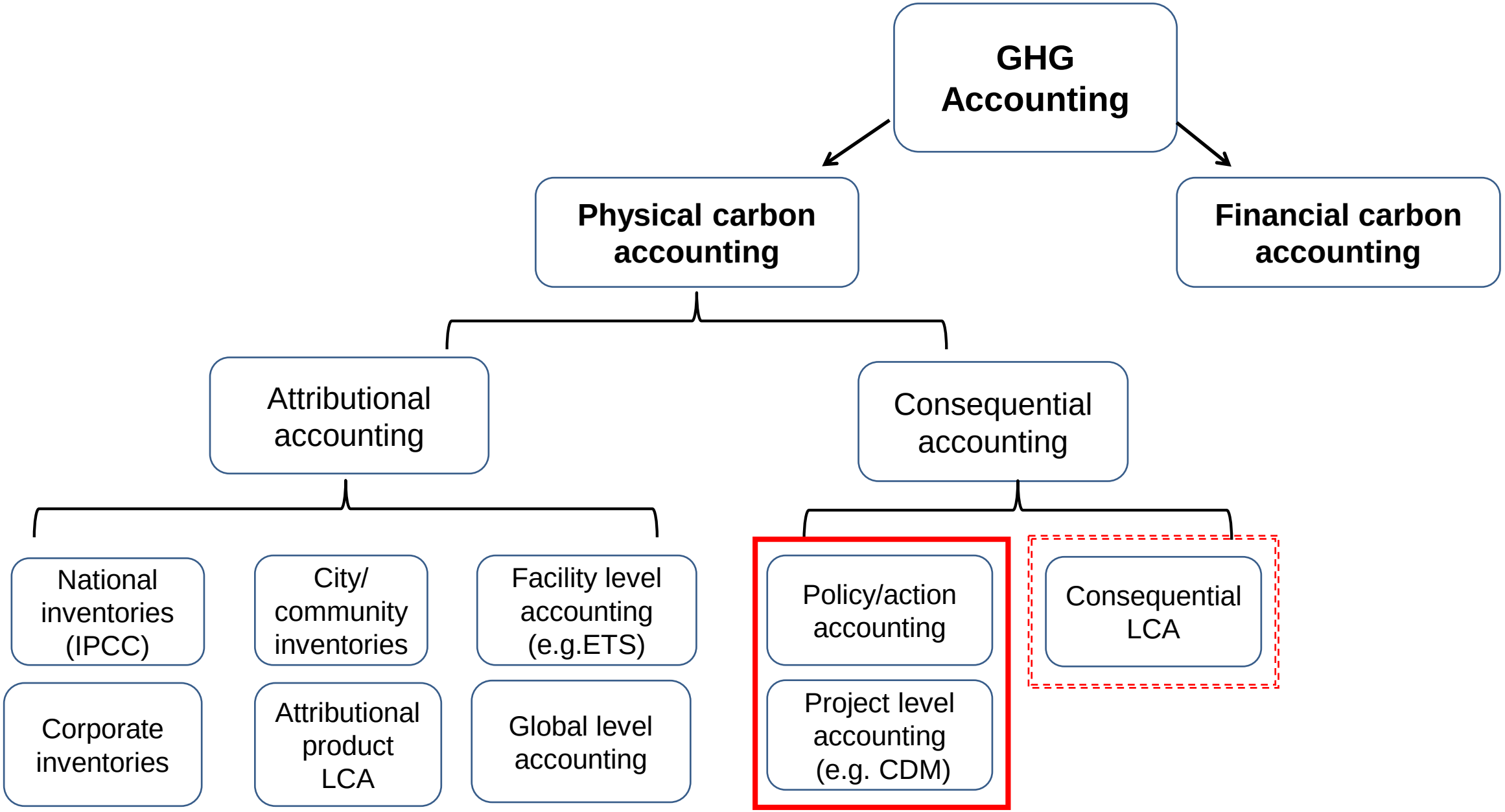
Consequences of increase in milk products:

avoided burden

↑ in production of milk → **↑ in supply of calves from dairy cows** → **↓ in demand for dedicated beef/pork** → **↓ in emissions**

Consequential life cycle assessment (*) - Standards

- Attributional life cycle assessment is still more common
  - No dedicated standard for consequential LCA
  - ISO 14044 and 14049 for (attributional) LCA
  - ILCD Handbook (micro, meso, and macro contexts): http://eplca.jrc.ec.europa.eu/?page_id=86
  - CALCAS guidelines: http://www.leidenuniv.nl/cml/ssp/publications/calcas_report_d18.pdf
  - Data bases – e.g. Ecoinvent 3 (by choosing different system models)

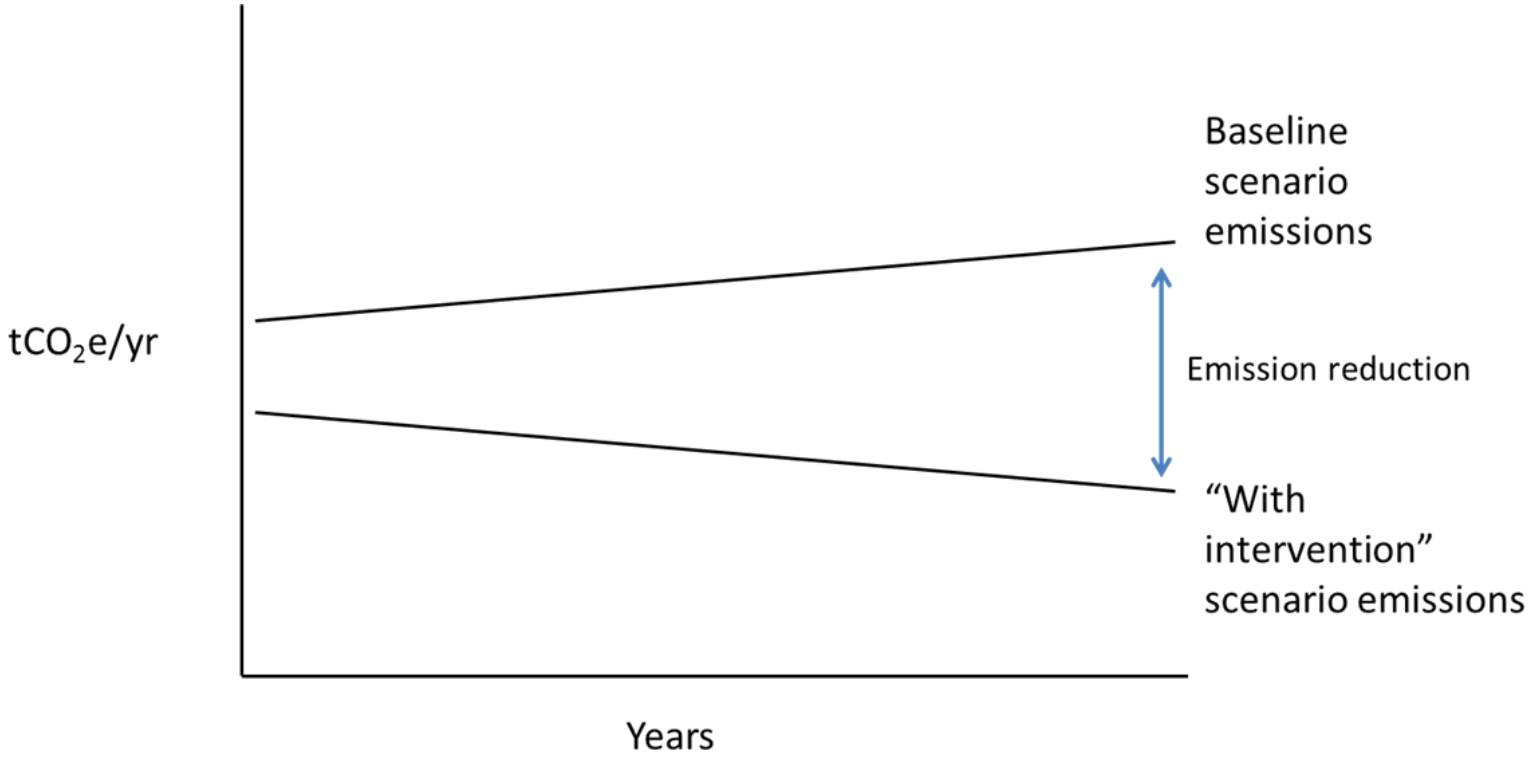

GHG Accounting
Physical carbon accounting
Financial carbon accounting
Attributional accounting
Consequential accounting
National inventories (IPCC)
City/ community inventories
Facility level accounting (e.g.ETS)
Corporate inventories
Attributional product LCA
Global level accounting
Policy/action accounting
Project level accounting (e.g. CDM)
Consequential LCA

02

Project/Policy Level Accounting

Project-level accounting – the concept

- Changes in emissions achieved by the implementation of a project



Project-level accounting – the standards

- ISO 14064-2 – See Canvas-Files
- GHG Protocol for Project accounting - https://ghgprotocol.org/project-protocol

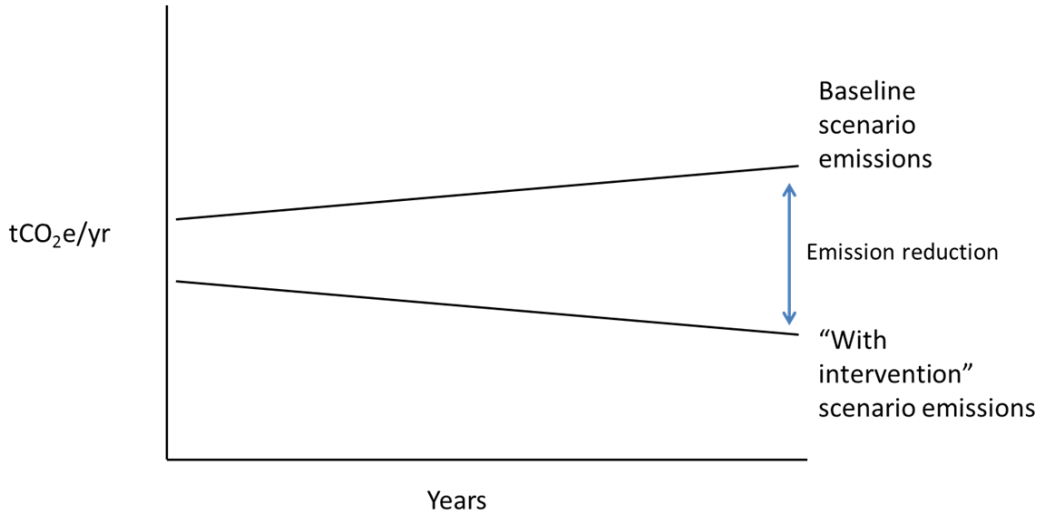
- Applications:
  - Internationally, CDM and VCS methodologies - http://www.cdmpipeline.org/cdm-methodologies.htm and http://www.v-c-s.org/methodologies/find
  - Domestically, China Certified Emission Reduction (CCER) methodologies https://ccer.cets.org.cn/notice/columnDetail?sign=methodology

Return to the wind farm case

A new offshore wind farm supplies electricity to the regional grid. The project developer claims:

"Our wind farm reduces CO_2 emissions because it generates electricity without burning fossil fuels."

- Using project accounting, what two scenarios should we compare?



Example 1 – What changes when the wind farm is built?

- A simplified example: electricity demand in the region is assumed to remain constant. The wind electricity is assumed to fully displace the coal-fired power.

Without the project



Electricity from an existing coal power plant.

Baseline scenario

With the project



Wind electricity replaces electricity generated from coal.

Project scenario

Where do the emission reductions come from?

Estimated emission reductions

$$ER = E_{baseline} - E_{project}$$

Assumptions: 1,000 MWh of wind electricity fully replaces coal generation; wind power has zero direct **operational** CO_2 emissions.
Emission factors of coal-generated electricity is assumed to be 0.9 tCO_2/MWh.

(Illustrative values only.)

$$\boldsymbol{ER = E_{baseline} - E_{project}}$$

Example 2 - Same Wind Farm. Different Emission Reductions?

- Imagine building the same wind farm in different locations.



Shanghai

45.3% of electricity supplied from outside the city (2024)
Local generation capacity remains largely thermal.



Xinjiang

64% of installed capacity from renewable energy (2025)
Major electricity exporter, with renewable energy integration challenges.

Would the two projects necessarily achieve the same GHG emission reductions? Why?

What electricity generation might each wind farm replace?

What if some wind electricity cannot be used because of curtailment?

What determines the estimated emission reductions from building a wind farm?

1. What generation is displaced?

marginal installation

   - Identify the affected electricity sources—not simply the regional average mix.

2. How much generation actually changes?

   - Consider wind availability, electricity use, curtailment, and transmission.

3. What evidence supports the estimate?

   - Check baseline assumptions, relevant data, assessment boundaries, and methodology.

Same installed capacity ≠ Same emission reductions

Project-level accounting – real-world example

减排量核算报告

项目名称	敦煌首航节能新能源有限公司敦煌 10 兆瓦熔盐塔式光热发电项目
项目登记编号	A110002
方法学及版本号	温室气体自愿减排项目方法学 并网光热发电（CCER—01—001—V01）
项目所属行业领域	能源产业（可再生/不可再生资源）
登记的项目设计文件版本 （适用于本核算期减排量核算报告）	3.0 版
减排量核算报告的版本号	1.0 版
减排量核算报告的完成日期	2024/12/03
核算期的顺序号	第一核算期
本核算期覆盖日期	2021/01/01 至 2024/12/31
申请项目登记的项目业主	敦煌首航节能新能源有限公司
项目所有者	敦煌首航节能新能源有限公司
项目设计文件中预先估算的本核算期内减排量	105,764 tCO_2e
本核算期内产生的减排量	31,192 tCO_2e

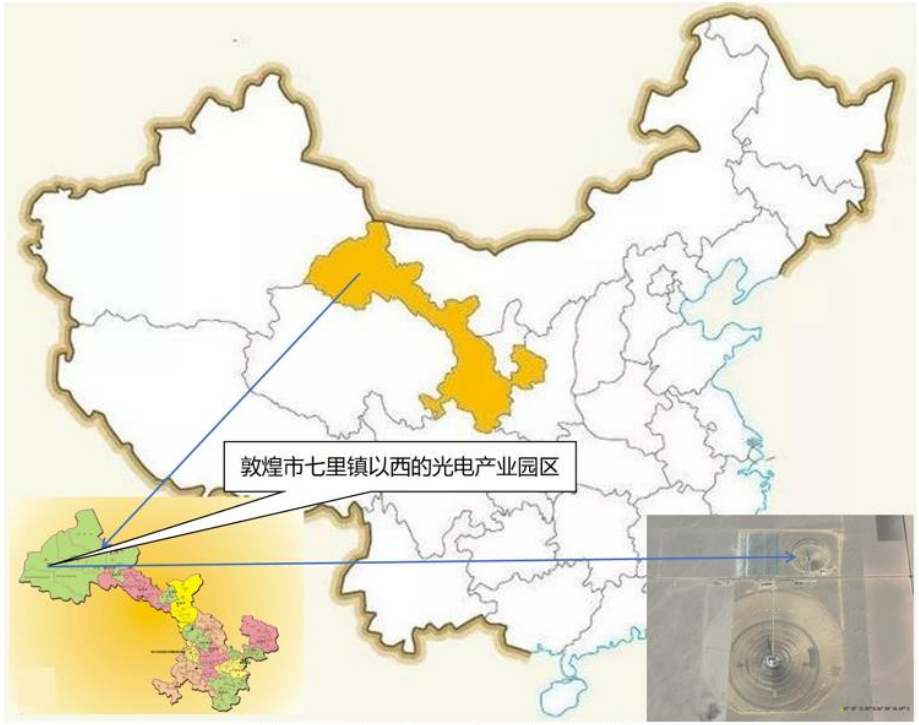


图 1：本项目在甘肃省及敦煌市地图上的位置

A.3. 所采用方法学

本项目采用《温室气体自愿减排项目方法学 并网光热发电》（CCER-01-001-V01）进行项目开发。



SJTU Goes Electric: Can You Help?

Scenario: SJTU plans to replace its diesel campus shuttle buses with electric buses.

The Campus Services Office invites student teams to help assess the GHG implications.

Your team receives two requests:

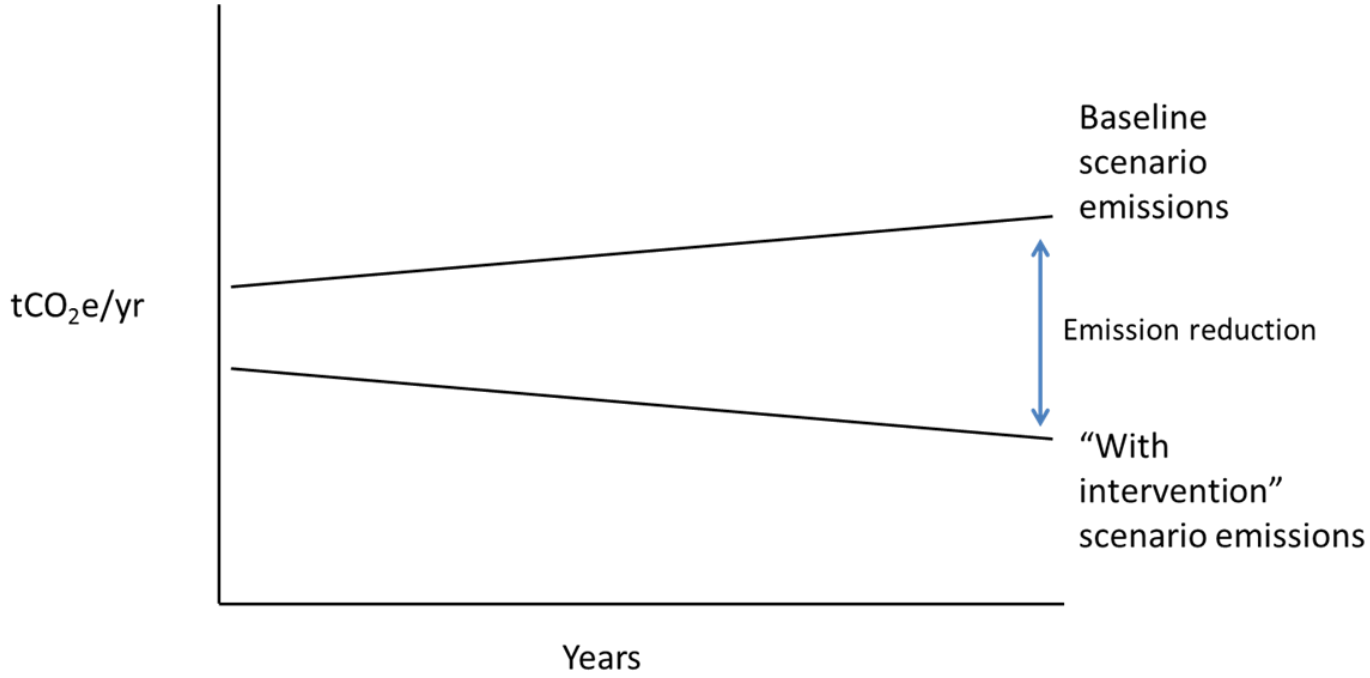
Task A — Estimate the annual GHG emissions associated with operating the campus shuttle fleet.

Task B — Estimate how much GHG emissions would change if diesel shuttles were replaced by electric buses.

Fill out the task sheet handed out in class, take a photo of your work, and post it on Yuketang.

Policy-level accounting

- The policy level accounting method is identical to the project level accounting method.



- Policy-level accounting standard:
  - GHG Protocol's Policy and Action Standard - https://ghgprotocol.org/policy-and-action-standard

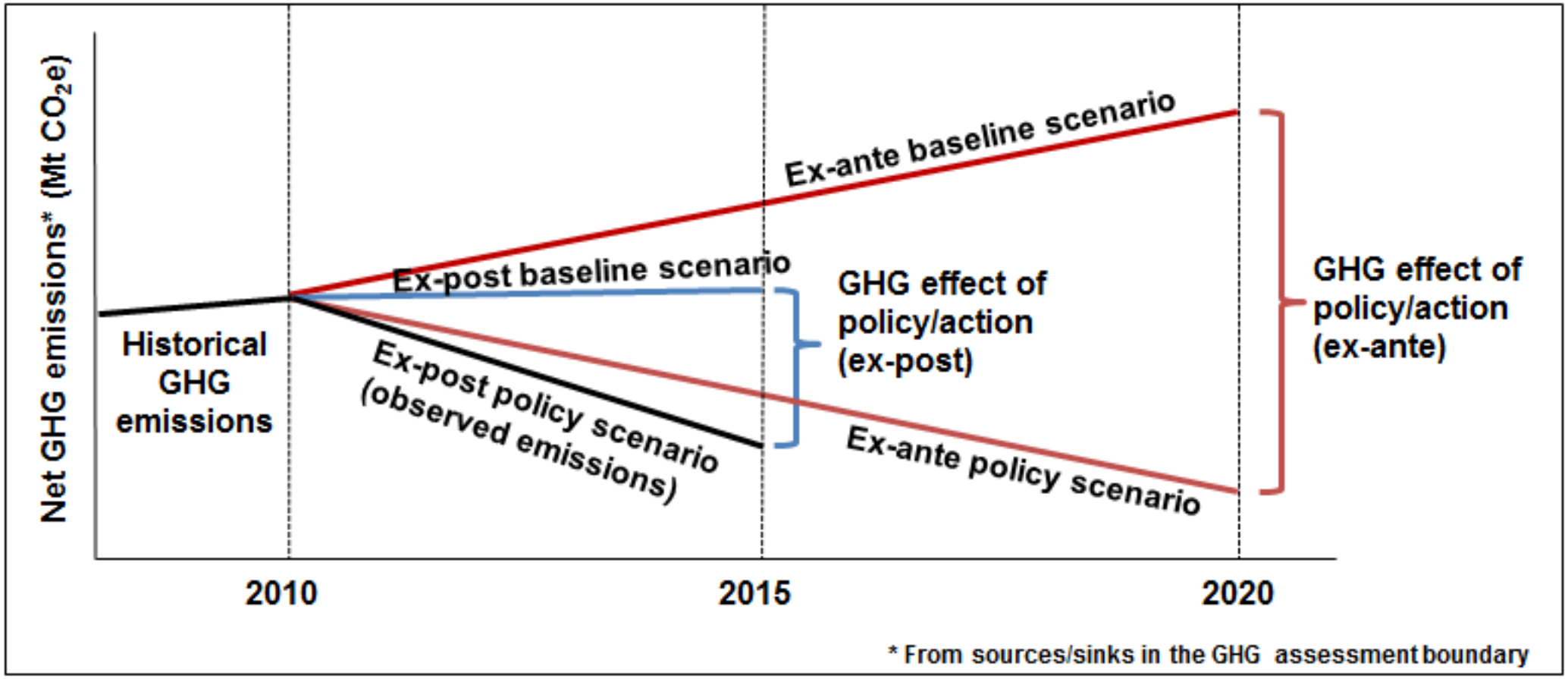
Table 1.2 **Comparison of the *Project Protocol* and the *Policy and Action Standard***

Standard	Applicability	Objectives	Differences in approach
Project Protocol	Individual mitigation projects, such as an individual solar photovoltaic installation	Focused primarily on crediting or offsetting	Provides detailed guidance on project-specific baselines, including addressing additionality of projects
Policy and Action Standard	Policies and actions at a larger scale than an individual project, such as renewable energy policies at the sectoral or jurisdiction level; policies and actions that increase or decrease emissions	Intended to support broader objectives (described in Chapter 2)	Provides guidance on estimating interactions between policies or actions; defining a baseline scenario at a larger scale than a project; and identifying and estimating various indirect effects at a broader scale, such as international leakage of emissions

Policy-level accounting – the standard

- GHG Protocol's Policy and Action standard https://ghgprotocol.org/policy-and-action-standard
  - It was published in 2014, which was the first-of-its-kind standard designed to help countries and local governments better design, track, and evaluate policies and actions.
    - Pilot tests were conducted on 27 policies and actions in 20 countries and cities.
  - It can be used at multiple points of time:
    - Before policy implementation (through ex-ante assessment)
    - During policy implementation
    - After policy implementation (through ex-post assesement)

Policy-level accounting – GHGP's Policy and Action Standard



(Source: https://ghgprotocol.org/sites/default/files/standards_supporting/Executive%20Summary%20%28English%29.pdf)

Policy-level accounting – the steps

Step 1 - define policy or action

Step 2 – identifying effects/mapping the causal chain

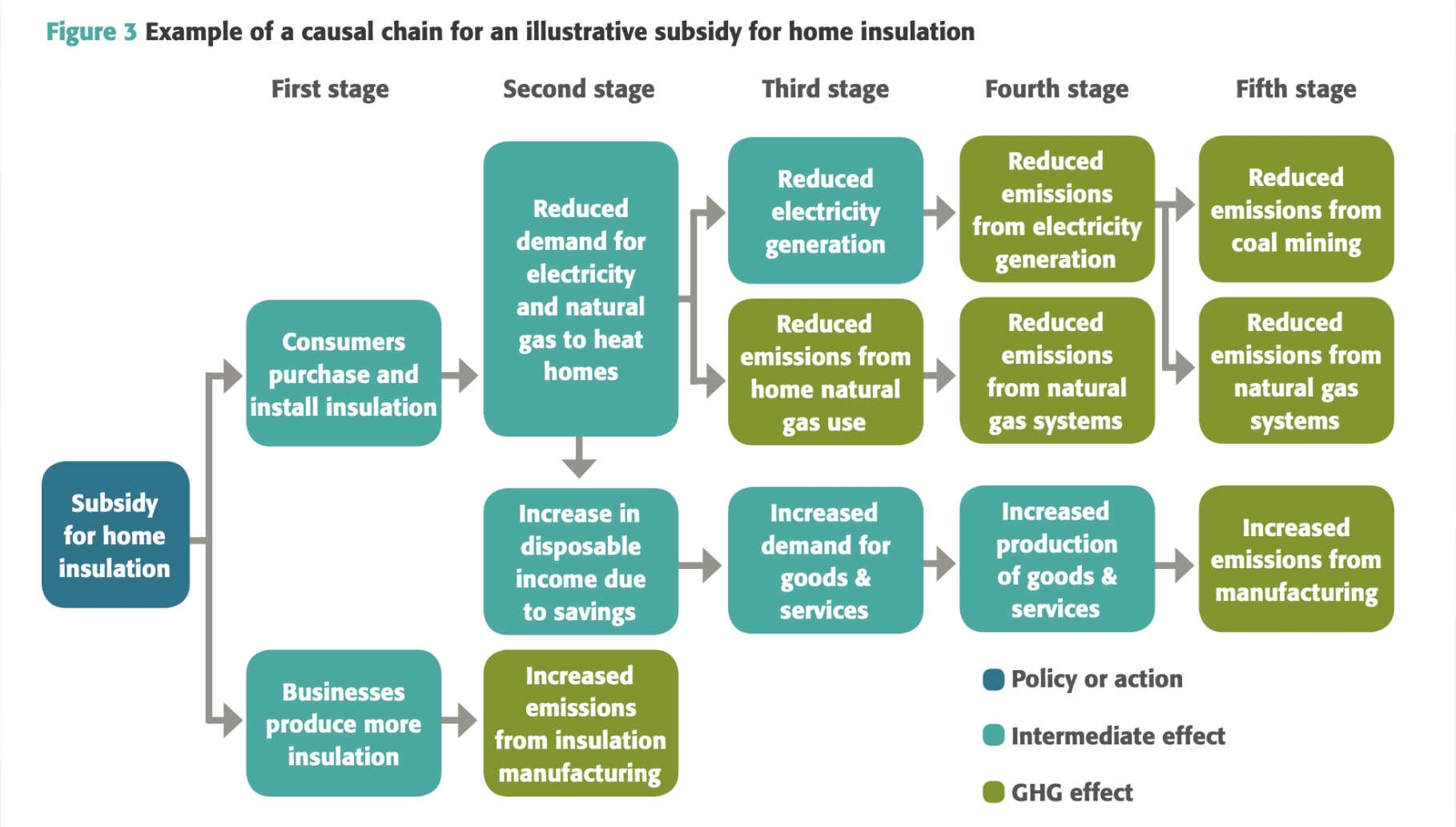
Step 3 – define the assessment boundary

Step 4 – calculate baseline emissions

Step 5 – calculate policy scenario emissions

Step 6 – “baseline emissions minus policy scenario emissions = change caused by policy”

Example of a causal chain for an illustrative subsidy for home insulation



(Source: https://ghgprotocol.org/sites/default/files/standards_supporting/Executive%20Summary%20%28English%29.pdf)

Carbon Accountin...

24.投票题 (单选)

投票 最多可选1项

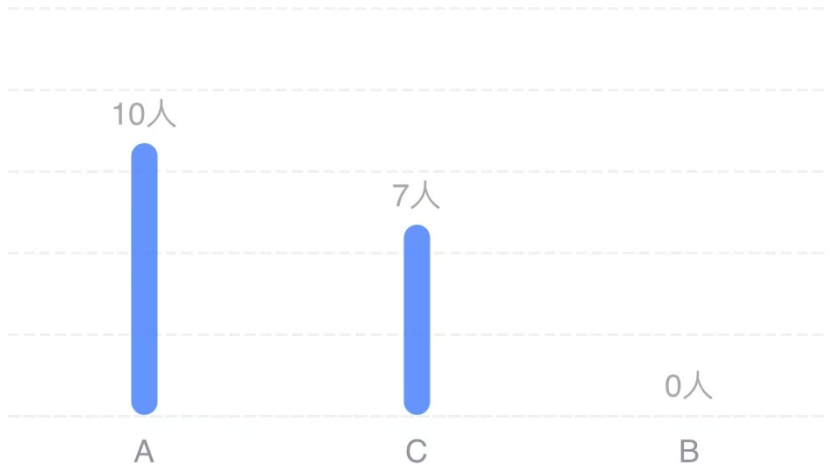
Do you think bioenergy is a carbon-reducing energy source?

A Yes.

B No.

C It depends.

9/24 Lecture 4



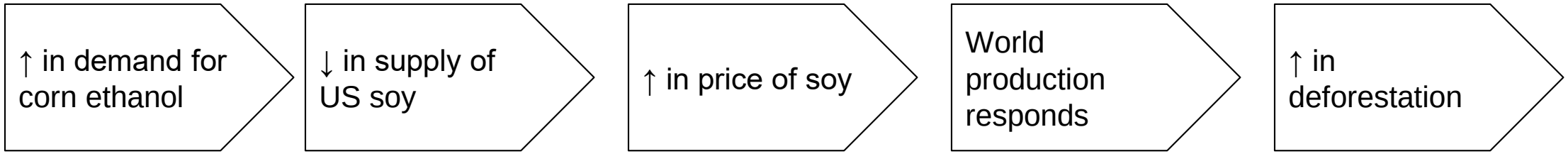
17人
参与投票

A
最多人选

Policy-level accounting – the example of biofuel policies

What assumptions might policy makers make when promoting biofuels as a 'green' solution?

- A brief summary of the key findings from Searchinger's 2008 paper:
  - Biofuels are promoted to reduce fossil fuel dependency.
  - Increased demand for biofuels can lead to agricultural land expansion potentially causing deforestation and higher net carbon emissions due to indirect land-use changes (ILUC).



Any critics?

主观题 10分

设置

Biofuel policies aim to reduce greenhouse gas emissions by replacing fossil fuels, but they often lead to indirect effects as Searchinger et al. (2008) argued.

Discuss in your group for 3 minutes:

- What indirect effects could arise from large-scale biofuel adoption?
- How should these potential unintended consequences influence biofuel policies?

Each group shares one insight or suggestion with the class.

作答

Source of fuel	Making feedstock	Refining fuel	Vehicle operation (burning fuel)	Net land-use effects: Feedstock carbon uptake from atmosphere (GREET)	Net land-use effects: Land-use change	Total GHGs	% Change in net GHGs versus gasoline
Gasoline	+4	+15	+72	0	–	+92	–
Corn ethanol (GREET)	+24	+40	+71	–62	–	+74 +135 without feedstock credit	–20% +47% without feedstock credit
Corn ethanol plus land use change	+24	+40	+71	–62	+104	+177	+93%

(Source:
Searchinger, T. et al., 2008. Use of U.S. croplands for biofuels increases greenhouse gases through emissions from land-use change. *Science*, 319(5867), pp.1238–40.)

03

Pros and Cons of Consequential Accounting

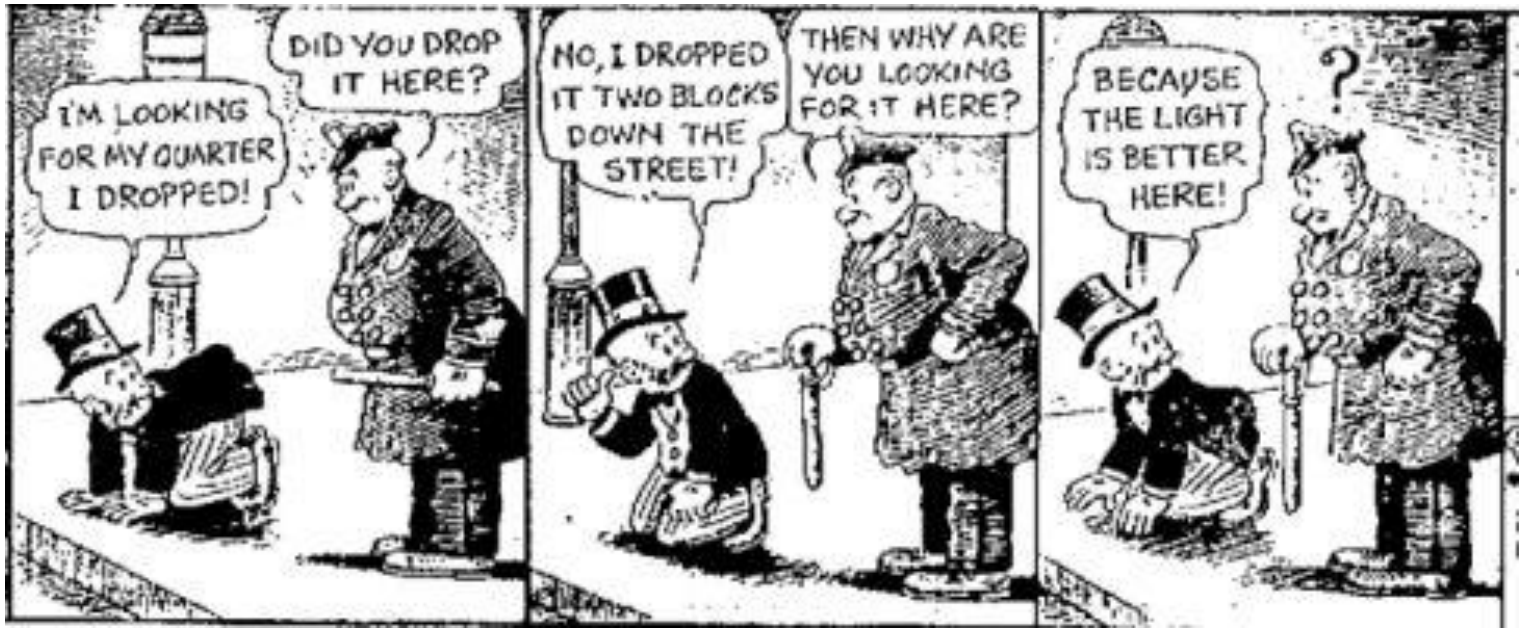
Pros with consequential accounting

- Arguably consequential methods are more relevant to decision making
  1. identifying opportunities to improve environmental performance;
  2. informing decision-makers for priority setting and process design;
  3. selecting indicators of environmental performance; and
  4. marketing

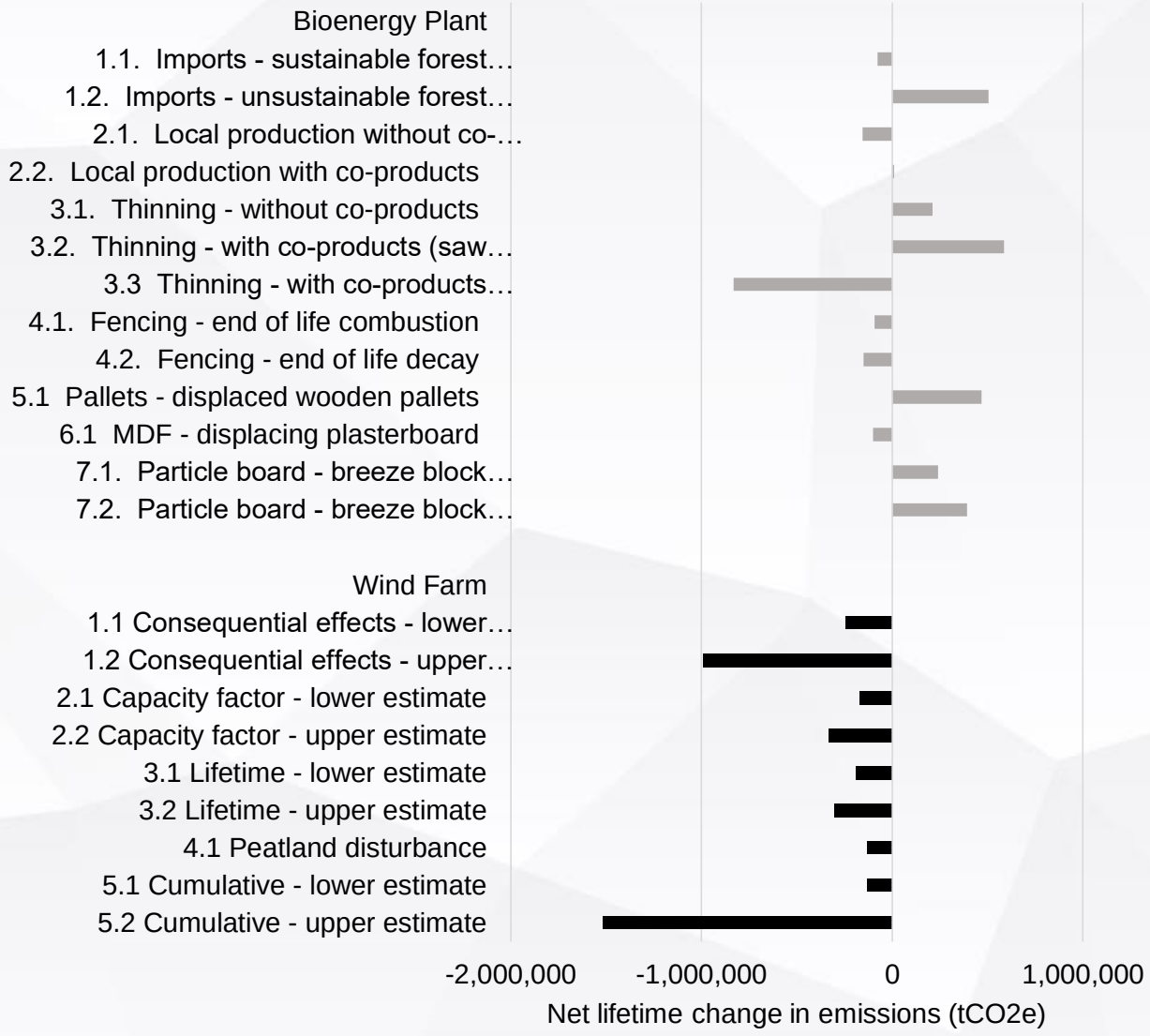
Cons with consequential accounting

- High uncertainty (but uncertainty is a finding sometimes)
- Difficult to implement

BUT – "streetlight" effect – what do we actually want to know?



Uncertainty is the finding - Example of bioheat plant and wind farm



From: Brander (2017)

投票 最多可选1项

设置

Which case makes you more uncomfortable in terms of mitigation if the analysis only uses attributional-style calculation? Give one reason.

Case A: Production decision
BMW increases recycled aluminum use across future vehicle models to lower product carbon footprints.

Case B: Project decision
An investor builds a new wind farm and claims carbon-reduction benefits.

A Product decision

B Project decision

C Both equally

D It depends on the claim being made

Think about:
- Who will use the results?
- What happens if the method gives the wrong signal?

提交

Product Decisions vs Project/Policy Decisions

- Consequential approach is theoretically relevant whenever a decision changes emissions. But its practical importance differs across decision contexts.

Product-level decisions	Project / policy-level claims
Often about product design, material choice, procurement, or communication	Often about investment, carbon credits, regulation, or policy evaluation
Consequential LCA may be theoretically appropriate	Consequential logic is central
But it is often time-consuming and data demanding	Baseline comparison is necessary for credibility
Attributional PCF is often used as a practical compromise	Inventory-style calculation can seriously mislead
Main risk: imperfect business decision	Main risk: wrong investment, non-real credits, misleading policy

Example: A new wind farm does not automatically create the same carbon-reduction benefit everywhere.

Appropriate use of attributional and consequential methods

Type of methodology	Purpose	Limitations
Attributional (inventory)	• Emissions/removals inventory • Assigning "responsibility" • Basis for cap and trade • Setting reduction targets – and tracking progress • "Hot spot" identification • Carbon budgets - additivity	• Does not show system-wide changes caused by decisions • Should not be used on its own to select mitigation actions
Consequential (assessment of total change)	• Informing selection of mitigation actions • Quantifying system-wide change in emissions caused by decision/action • Quantification for tradable emission reduction credits	• Does not show absolute anthropogenic emissions to the atmosphere • Complex to implement

Appropriate use of attributional and consequential methods

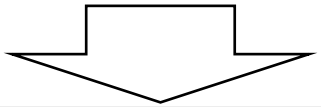
- When managing an attributional inventory - need to use consequential methods to appraise mitigation options :
  - To avoid leakage/unintended impacts elsewhere in the system
    - E.g. Searchinger et al (2008) on biofuel policy
  - To report "enabled reductions" not captured in the attributional inventory
  - Calculate marginal abatement cost curves ($/tCO$_2$e abated)

- Attributional and consequential methods can compliment each other

Step 1. Attributional accounting method

Purpose:

a. Define initial scope of responsibility (set of impacts for business to manage)
b. Set targets for reducing impacts (and measure progress over time)



Step 2. Consequential accounting method

Purpose:

a. Check system-wide consequences of actions aimed at reducing impacts

04

Wrap Up and Next Time

Wrap up

Consequential methods aim to measure total change in emissions as a result of a decision/intervention

Three main types of consequential assessment:

- Consequential LCA (methodological origin)
- Project-level accounting
- Policy-level accounting

Pros and cons:

- Consequential methods are suited to decision making
- More difficult to implement, and higher uncertainty (BUT they aim to tell us what we want to know)

Next Time – Financial Carbon Accounting

- Essential reading:
  - Task Force on Climate-related Financial Disclosures (TCFD). Final Report: Recommendations of the Task Force on Climate-related Financial Disclosures. June 29, 2017.(Read through Section B) https://assets.bbhub.io/company/sites/60/2020/10/FINAL-2017-TCFD-Report-11052018.pdf
  - PCAF. The Global GHG Accounting & Reporting Standard for the Financial Industry[R]. 2020 (Browse Chapter 4 and Read through page 42 to 46) https://carbonaccountingfinancials.com/standard

- Recommended reading
  - 饶淑玲. 2022). 国外金融机构碳核算的经验. 中国金融, 2022(02), 88-89.





See you next time